Complete Problem Index — by Template Category

All **737 problems** from `LeetCode_Complete_Reference.tex`, assigned to their template category. ★ marks problems with a worked template/solution already in this file's category sections above; unmarked problems are catalogued here for coverage. Source category shown when it differs (e.g. Trees & Backtracking → DFS; Heap → Stack/Queue).

Two Pointers (93 problems, 93 templated)

#	★	Name	Complexity	Description
1	★	Two Sum	O(n) / O(1)	Given an integer array and a target sum, return the indices of two numbers that add up to target. Exactly one solution exists; cannot use the same element twice.
11	★	Container With Most Water	O(n) / O(1)	Given n vertical lines at positions 1..n with heights, find two lines that together with the x-axis form a container holding the most water.
15	★	3Sum	O(n^2) / O(n)	Find all unique triplets [a,b,c] in an integer array such that a+b+c=0. No duplicate triplets.
16	★	3Sum Closest		Find three integers closest to a given target sum. Return the sum of those three integers.
18	★	4Sum	O(n^3) / O(n)	Find all unique quadruplets in an array that sum to a given target. No duplicates.
26	★	Remove Duplicates from Sorted Array	O(n) / O(1)	Remove duplicates from a sorted array in-place. Return the count of unique elements.
27	★	Remove Element	O(n) / O(1)	Remove all occurrences of a value from an array in-place. Return the new length.
31	★	Next Permutation		Rearrange an array of integers to its lexicographically next greater permutation. If none, rearrange to lowest order.
41	★	First issing Positive		Find the smallest missing positive integer in an unsorted array. Must run in O(n) time and O(1) space.
42	★	Trapping Rain Water	O(n) / O(1)	Given an elevation map (array of bar heights), compute how much water it can trap after raining.
48	★	Rotate Image		Rotate an n×n matrix 90 degrees clockwise in-place.
54	★	Spiral Matrix		Return all elements of an m×n matrix in spiral order.
73	★	Set Matrix Zeroes		Given an m×n matrix, set entire row and column to 0 if any element is 0. Do it in-place.
75	★	Sort Colors	O(n) / O(1)	Sort an array of 0s, 1s, and 2s in-place without using sort (Dutch National Flag problem).
80	★	Remove Duplicates from Sorted Array II	O(n) / O(1)	Remove duplicates from a sorted array so each element appears at most twice in-place.
88	★	Merge Sorted Array	O(m+n) / O(1)	Merge two sorted integer arrays nums1 and nums2 into nums1 in-place (nums1 has extra space).
128	★	Longest Consecutive Sequence		Find the length of the longest consecutive sequence of integers in an unsorted array. O(n) required.
164	★	Maximum Gap	O(n) / O(n)	Find the maximum gap between successive elements in a sorted form of an array. O(n) required.
167	★	Two Sum II - Input Array Is Sorted		Given a sorted array, find two numbers that add to a target (1-indexed). Exactly one solution.
169	★	Majority Element		Find the majority element that appears more than n/2 times.
189	★	Rotate Array		Rotate an array to the right by k positions in-place.
217	★	Contains Duplicate		Check if any value appears at least twice in an integer array.
219	★	Contains Duplicate II	O(n) / O(k)	Check if there are two distinct indices i and j such that nums[i]==nums[j] and i-j.
220	★	Contains Duplicate III		Check if there are two indices i and j such that nums[i]-nums[j] and i-j.
229	★	Majority Element II		Find all elements appearing more than n/3 times in an array.
238	★	Product of Array Except Self		Return an array where output[i] is the product of all elements except nums[i]. No division.

#	★	Name	Complexity	Description
259	★	3Sum Smaller		Count triplets with sum smaller than a target (sorted array).
280	★	Wiggle Sort		Rearrange an array in-place so it alternates peaks and valleys.
283	★	Move Zeroes	O(n) / O(1)	Move all zeroes in an array to the end while maintaining relative order of non-zero elements.
289	★	Game of Life		Simulate Conway's Game of Life: update all cells simultaneously based on neighbor counts.
303	★	Range Sum Query - Immutable		Answer multiple range sum queries on a static array efficiently.
304	★	Range Sum Query 2D - Immutable		Answer multiple 2D region sum queries on a static matrix efficiently.
307	★	Range Sum Query - Mutable		Support range sum queries and point updates on an array.
311	★	Sparse Matrix Multiplication		Multiply two sparse matrices efficiently (skip zero elements).
315	★	Count of Smaller Numbers After Self	O(n log n) / O(n)	For each element, count elements to its right that are smaller.
325	★	Maximum Size Subarray Sum Equals k	O(n) / O(n)	Find the maximum length subarray with sum equal to k.
327	★	Count of Range Sum	O(n log n) / O(n)	Count range sums that lie in [lower, upper].
334	★	Increasing Triplet Subsequence		Return true if there exists `i < j < k` with `nums[i] < nums[j] < nums[k]`.
349	★	Intersection of Two Arrays		Find the intersection of two integer arrays (unique elements only).
350	★	Intersection of Two Arrays II		Find the intersection of two integer arrays including duplicates.
370	★	Range Addition		Apply range updates `[start, end] += inc` then return the final array.
384	★	Shuffle an Array		Shuffle an integer array so each permutation is equally likely.
414	★	Third Maximum Number		Find the third maximum distinct integer. Return the maximum if no third.
419	★	Battleships in a Board		Count battleships in a board ('X's forming horizontal or vertical lines).
442	★	Find All Duplicates in an Array		Values in [1,n], each appears once or twice. Return the ones appearing twice. O(n) time, O(1) extra space.
448	★	Find All Numbers Disappeared in an Array		Find all integers in range [1,n] that are missing from an array of size n.
454	★	4Sum II		Count quadruplets (i,j,k,l) such that A[i]+B[j]+C[k]+D[l]==0.
457	★	Circular Array Loop		Detect a cycle of length > 1 with consistent direction in a circular array of jumps.
485	★	Max Consecutive Ones		Find the maximum number of consecutive 1s in a binary array.
493	★	Reverse Pairs		Count pairs (i,j) with `i < j` and `nums[i] > 2 * nums[j]`.
498	★	Diagonal Traverse		Traverse an m×n matrix diagonally, alternating direction.
517	★	Super Washing Machines		Minimize the max number of moves to equalize dresses across machines (one dress to a neighbor per move).
523	★	Continuous Subarray Sum	O(n) / O(n)	Check if a subarray exists with sum divisible by k.
525	★	Contiguous Array		Find the maximum length subarray with equal number of 0s and 1s.
532	★	K-diff Pairs in an Array		Count unique k-diff pairs (i,j) in an array where nums[i]-nums[j]==k.
560	★	Subarray Sum Equals K	O(n) / O(n)	Count subarrays that sum to exactly k.
566	★	Reshape the Matrix		Reshape an m×n matrix to r×c without changing element order.
581	★	Shortest Unsorted Continuous Subarray		Find the shortest subarray that, if sorted, makes the whole array sorted.

#	★	Name	Complexity	Description
594	★	Longest Harmonious Subsequence		Find the longest subsequence where max and min differ by exactly 1.
599	★	Minimum Index Sum of Two Lists		Find the common strings between two lists with the minimum index sum.
611	★	Valid Triangle Number		Count the number of valid triangles from a sorted array of lengths.
661	★	Image Smoother		Replace each cell with the floor of the average of itself and its up-to-8 neighbors.
723	★	Candy Crush		Simulate Candy Crush: repeatedly remove groups of 3 same-color candies and drop.
724	★	Find Pivot Index		Find a pivot index where left sum equals right sum.
766	★	Toeplitz Matrix		Check if a matrix is Toeplitz (all diagonals have the same value).
769	★	Max Chunks To Make Sorted		Array is a permutation of [0,n-1]. Max number of chunks that can be sorted independently to sort the whole.
795	★	Number of Subarrays with Bounded Maximum		Count subarrays with a maximum value in range [L, R].
807	★	Max Increase to Keep City Skyline		Maximize the total increase in building heights while preserving the city's skyline from all 4 directions.
825	★	Friends Of Appropriate Ages		Count pairs of people where one person's age allows them to friend the other.
838	★	Push Dominoes		Given a string of dominoes ('L', 'R', '.'), simulate their fall.
845	★	Longest Mountain in Array		Find the length of the longest mountain subarray (increases then decreases).
896	★	Monotonic Array		Check if an array is monotonically increasing or decreasing.
912	★	Sort an Array	O(n log n) / O(n)	Sort an integer array. Implement an O(n log n) sort.
974	★	Subarray Sums Divisible by K	O(n) / O(k)	Count subarrays whose sum is divisible by k.
977	★	Squares of a Sorted Array		Return the squares of a sorted array in sorted order.
1013	★	Partition Array Into Three Parts With Equal Sum		Can the array be split into three contiguous parts with equal sums?
1074	★	Number of Submatrices That Sum to Target		Count submatrices summing to target.
1109	★	Corporate Flight Bookings		Given bookings `[first, last, seats]`, return total seats booked per flight.
1213	★	Intersection of Three Sorted Arrays		Find the intersection of three sorted arrays.
1275	★	Find Winner on a Tic Tac Toe Game		Determine the winner of a Tic-Tac-Toe game given a sequence of moves.
1351	★	Count Negative Numbers in a Sorted Matrix		Count the number of negative numbers in a sorted matrix.
1424	★	Diagonal Traverse II		Traverse a matrix diagonally from bottom-left to top-right.
1460	★	Make Two Arrays Equal by Reversing Sub-arrays		Check if two arrays become equal after reversing any sub-array of one.
1480	★	Running Sum of 1d Array		Return the running sum of an array.
1498	★	Number of Subsequences That Satisfy the Given Sum Condition		Count the number of subsequences where min+max target.
1748	★	Sum of Unique Elements		Find the sum of all unique elements in an array.
1762	★	Buildings With an Ocean View		Find all buildings with an ocean view (nothing taller to their right).
1868	★	Product of Two Run-Length Encoded Arrays		Multiply two run-length encoded arrays.
1877	★	Minimize Maximum Pair Sum in Array		Minimize the maximum pair sum after optimally pairing elements.

#	★	Name	Complexity	Description
1893	★	Check if All the Integers in a Range Are Covered		Check if all integers in [left, right] are covered by at least one given range.
1894	★	Find the Student that Will Replace the Chalk		Find the index of the student who will replace the chalk.
1920	★	Build Array from Permutation		Build an array where result[i] = nums[nums[i]].
1929	★	Concatenation of Array		Return the concatenation of an array with itself.

Sliding Window (21 problems, 21 templated)

#	★	Name	Complexity	Description
3	★	Longest Substring Without Repeating Characters	O(n) / O(1)	Find the length of the longest substring without any repeating characters in a given string.
30	★	Substring with Concatenation of All Words		Find all starting indices in s where a concatenation of all words in an array appears as a substring.
76	★	Minimum Window Substring	O(n) / O(k)	Find the minimum window substring of s that contains all characters in t.
159	★	Longest Substring with At Most Two Distinct Characters	O(n) / O(1)	Find the length of the longest substring containing at most two distinct characters.
187	★	Repeated DNA Sequences		Find all 10-letter DNA sequences (substrings) that occur more than once.
209	★	Minimum Size Subarray Sum		Find the minimal length subarray whose sum target. Return 0 if none.
239	★	Sliding Window Maximum	O(n) / O(k)	Return the maximum value in each sliding window of size k across an array.
340	★	Longest Substring with At Most K Distinct Characters	O(n) / O(k)	Find the length of the longest substring containing at most k distinct characters.
395	★	Longest Substring with At Least K Repeating Characters		Find the longest substring where every character appears at least k times.
424	★	Longest Repeating Character Replacement	O(n) / O(1)	Find the length of the longest substring where you can replace at most k characters to make all same.
438	★	Find All Anagrams in a String	O(n) / O(1)	Find all anagram start indices of pattern p in string s.
487	★	Max Consecutive Ones II		Given a binary array `nums`, return the maximum number of consecutive 1s if you may flip at most one 0.
567	★	Permutation in String	O(n) / O(1)	Check if any permutation of pattern p exists as a substring of string s.
643	★	Maximum Average Subarray I		Find the maximum average of any subarray of length k.
689	★	Maximum Sum of 3 Non-Overlapping Subarrays		Find 3 non-overlapping subarrays of length k with maximum total sum.
713	★	Subarray Product Less Than K		Return the number of contiguous subarrays whose product of all elements is strictly less than `k`.
727	★	Minimum Window Subsequence		Find the minimum window in s that contains all characters of t in order (subsequence).
1004	★	Max Consecutive Ones III	O(n) / O(1)	Find the maximum consecutive 1s if you can flip at most k zeros.
1044	★	Longest Duplicate Substring		Find the longest duplicate substring using binary search and rolling hash.
1343	★	#1343	O(n) / O(1)	Count subarrays of size k with average >= threshold.
1838	★	Frequency of the Most Frequent Element		Find the minimum operations to make an array element appear the most frequently in a window.

Binary Search (30 problems, 30 templated)

#	★	Name	Complexity	Description
4	★	Median of Two Sorted Arrays		Find the median of two sorted arrays of sizes m and n in O(log(m+n)) time.
33	★	Search in Rotated Sorted Array	O(log n) / O(1)	Search for a target in a sorted array that has been rotated at an unknown pivot. Return index or -1.
34	★	Find First and Last Position of Element in Sorted Array	O(log n) / O(1)	Find the starting and ending positions of a target value in a sorted array. Return [-1,-1] if not found.
35	★	Search Insert Position	O(log n) / O(1)	Find the index where a target would be inserted in a sorted array to keep it sorted.
69	★	Sqrt(x)		Compute the integer square root of a non-negative integer x without using sqrt().
74	★	Search a 2D Matrix		Search for a target in an m×n matrix where each row is sorted and each row's first element > previous row's last.
81	★	Search in Rotated Sorted Array II		Search in a sorted, rotated array that may contain duplicates.
153	★	Find Minimum in Rotated Sorted Array		Find the minimum element in a sorted, rotated array with no duplicates.
162	★	Find Peak Element	O(log n) / O(1)	Find any peak element (element greater than its neighbors) in an array. O(log n) required.
240	★	Search a 2D Matrix II		Search for a target in an m×n matrix where rows and columns are both sorted.
278	★	First Bad Version		Find the first bad version using a provided isBadVersion(n) API. Minimize API calls.
374	★	Guess Number Higher or Lower		Guess the number between 1 and n using a provided guess() API. Minimize calls.
378	★	Kth Smallest Element in a Sorted Matrix	O(n log n) / O(n)	Find the kth smallest element in an n×n matrix where rows and columns are sorted.
410	★	Split Array Largest Sum	O(n log sum) / O(1)	Split an array into m subarrays to minimize the largest subarray sum.
475	★	Heaters		Given house and heater positions, find the minimum heater radius so every house is covered by some heater.
540	★	Single Element in a Sorted Array		Find the single non-duplicate element in a sorted array where all others appear twice. O(log n).
658	★	Find K Closest Elements		Find k integers in a sorted array closest to x, ordered by closeness then value.
704	★	Binary Search	O(log n) / O(1)	Classic binary search: find target in sorted array.
719	★	Find K-th Smallest Pair Distance		Find the k-th smallest absolute distance among all pairs in the array.
852	★	Peak Index in a Mountain Array		Find the peak index in a mountain array (element greater than both neighbors).
875	★	Koko Eating Bananas	O(n log max) / O(1)	Find the minimum eating speed k so Koko can eat all banana piles within h hours.
1011	★	Capacity To Ship Packages Within D Days	O(n log sum) / O(1)	Find the minimum capacity to ship all packages within D days.
1060	★	Missing Element in Sorted Array		Find the kth missing number in a sorted array.
1231	★	#1231	O(n log max) / O(1)	Max min-piece when cutting chocolate into k pieces.
1283	★	#1283	O(n log max) / O(1)	Smallest divisor so sum of ceil(nums[i]/divisor) <= threshold.

#	★	Name	Complexity	Description
1428	★	Leftmost Column with at Least a One		Find the leftmost column with at least one '1' in a binary matrix.
1482	★	Minimum Number of Days to Make m Bouquets		Find the minimum number of days to make m bouquets of k adjacent flowers.
1539	★	Kth Missing Positive Number		Find the kth missing positive integer.
1818	★	Minimum Absolute Sum Difference		Find the minimum absolute sum difference after one substitution.
1891	★	Cutting Ribbons		Find the maximum ribbon length such that m ribbons of that length can be cut.

Linked List (31 problems, 31 templated)

#	★	Name	Complexity	Description
2	★	Add Two Numbers	O(n) / O(1)	Add two non-negative integers represented as reversed linked lists (ones digit first). Return the sum as a reversed linked list.
19	★	Remove Nth Node From End of List		Remove the nth node from the end of a linked list and return the head.
21	★	Merge Two Sorted Lists	O(n) / O(1)	Merge two sorted linked lists into one sorted linked list in-place.
24	★	Swap Nodes in Pairs		Swap every two adjacent nodes in a linked list and return the modified head.
25	★	Reverse Nodes in k-Group	O(n) / O(1)	Reverse nodes in a linked list k at a time. If remaining nodes < k, leave them as-is.
61	★	Rotate List		Rotate a linked list to the right by k places.
82	★	Remove Duplicates from Sorted List II	O(n) / O(1)	Remove all nodes with duplicate values from a sorted linked list; keep only distinct values.
83	★	Remove Duplicates from Sorted List		Remove duplicate nodes from a sorted linked list; keep one copy of each value.
86	★	Partition List		Partition a linked list so all nodes with value < x come before nodes with value x.
92	★	Reverse Linked List II		Reverse nodes m through n (1-indexed) of a linked list in one pass.
109	★	Convert Sorted List to Binary Search Tree		Convert a sorted linked list to a height-balanced BST.
138	★	Copy List with Random Pointer		Deep-copy a linked list where each node has a 'random' pointer to any node or null.
141	★	Linked List Cycle	O(n) / O(1)	Detect if a linked list has a cycle.
142	★	Linked List Cycle II	O(n) / O(1)	Find the node where a cycle begins in a linked list.
143	★	Reorder List		Reorder a linked list L01… to L01-1… in-place.
148	★	Sort List		Sort a linked list in O(n log n) time and O(1) extra space.
160	★	Intersection of Two Linked Lists	O(n) / O(1)	Find the node where two singly linked lists intersect.
203	★	Remove Linked List Elements	O(n) / O(1)	Remove all nodes with a given value from a linked list.
206	★	Reverse Linked List		Reverse a singly linked list.
234	★	Palindrome Linked List	O(n) / O(1)	Check if a linked list is a palindrome in O(n) time and O(1) space.
237	★	Delete Node in a Linked List		Delete a node from a singly linked list given only a reference to that node (not the head).
287	★	Find the Duplicate Number	O(n) / O(1)	Find the duplicate number in an array of n+1 integers in range [1,n]. O(1) space required.
328	★	Odd Even Linked List		Rearrange a linked list so all odd-indexed nodes come before even-indexed nodes.
369	★	Plus One Linked List		A non-negative integer is stored as a linked list of digits in big-endian order (most significant first). Add one to it and return the resulting list.
430	★	Flatten a Multilevel Doubly Linked List		Flatten a multilevel doubly linked list where nodes may have a child list.
445	★	Add Two Numbers II		Two numbers are stored as linked lists in big-endian order (most significant digit first). Add them and return the sum as a linked list, also in big-endian order.
708	★	Insert into a Sorted Circular Linked List		Insert a value into a sorted circular linked list.
725	★	Split Linked List in Parts		Split a linked list into k consecutive parts as equal in size as possible.
876	★	Middle of the Linked List	O(n) / O(1)	Find the middle node of a linked list (if two middles, return second).

#	★	Name	Complexity	Description
1171	★	Remove Zero Sum Consecutive Nodes from Linked List		Remove consecutive nodes from a linked list whose values sum to zero.
1265	★	Print Immutable Linked List in Reverse		Print all values of an immutable linked list in reverse without modifying it.

String (87 problems, 87 templated)

#	★	Name	Complexity	Description
6	★	ZigZag Conversion		Convert a string into a zigzag pattern across numRows rows and read it row by row.
7	★	Reverse Integer	O(log n) / O(1)	Reverse the digits of a 32-bit signed integer. Return 0 if the reversed value overflows.
8	★	String to Integer (atoi)		Implement atoi: parse a string, skip whitespace, handle optional sign, read digits, clamp to 32-bit integer range.
12	★	Integer to Roman		Convert an integer to its Roman numeral representation (values 1--3999).
13	★	Roman to Integer		Convert a Roman numeral string to an integer.
14	★	Longest Common Prefix		Find the longest common prefix string among an array of strings. Return '' if none.
28	★	Implement strStr()		Find the first occurrence of needle in haystack. Return -1 if not found.
38	★	Count and Say		Generate the nth term of the 'count and say' sequence: read digits of previous term aloud.
43	★	Multiply Strings		Multiply two non-negative integers given as strings. Return product as a string.
49	★	Group Anagrams	O(nk) / O(n)	Group an array of strings so that anagrams are together.
58	★	Length of Last Word		Return the length of the last word in a string (word = max sequence of non-space chars).
65	★	Valid Number		Validate whether a string is a valid decimal number (integers, fractions, exponents).
67	★	Add Binary	O(n) / O(n)	Add two binary strings and return their binary sum as a string.
68	★	Text Justification		Format words into lines of maxWidth, fully justified (equal spaces, last line left-aligned).
125	★	Valid Palindrome		Check if a string is a palindrome, ignoring non-alphanumeric characters and case.
151	★	Reverse Words in a String		Reverse the order of words in a string (split on spaces, handle multiple spaces).
161	★	One Edit Distance		Determine if two strings are one edit distance apart (insert, delete, or replace one char).
163	★	Missing Ranges		Given a sorted array, return missing ranges between lower and upper bounds.
165	★	Compare Version Numbers		Compare two version number strings (dot-separated integers). Return -1, 0, or 1.
166	★	Fraction to Recurring Decimal		Convert a fraction numerator/denominator to a decimal string, noting repeating parts in parentheses.
168	★	Excel Sheet Column Title	O(log n) / O(1)	Convert an integer to its Excel column title (1, 26, 27, …).
171	★	Excel Sheet Column Number	O(L) / O(1)	Convert an Excel column title (A1, Z26, AA27) to its integer number.
179	★	Largest Number		Arrange numbers so their concatenation forms the largest possible number.
186	★	Reverse Words in a String II		Reverse words in a character array in-place (words separated by spaces).
205	★	Isomorphic Strings		Check if two strings follow the same character-substitution pattern (isomorphic).
214	★	Shortest Palindrome		Find the shortest palindrome by adding characters in front of a string.
228	★	Summary Ranges		Given a sorted unique integer array, summarize consecutive runs as ranges.
242	★	Valid Anagram	O(n) / O(1)	Determine if two strings are anagrams of each other.
246	★	Strobogrammatic Number		Check if a number reads the same when rotated 180 degrees.
249	★	Group Shifted Strings		Group strings that follow the same shift sequence (each char shifted by the same amount).

#	★	Name	Complexity	Description
266	★	Palindrome Permutation		Check if a string can be rearranged into a palindrome (at most one odd-count char).
271	★	Encode and Decode Strings		Encode a list of strings to a single string and decode it back.
273	★	Integer to English Words		Convert a non-negative integer to its English words representation.
290	★	Word Pattern		Check if a string s follows a given pattern (bijective character mapping).
299	★	Bulls and Cows		Count bulls (right digit, right place) and cows (right digit, wrong place) in a number guessing game.
344	★	Reverse String		Reverse a character array in-place.
345	★	Reverse Vowels of a String		Reverse only the vowels of a string.
383	★	Ransom Note	O(n) / O(1)	Determine if a ransom note can be constructed from a magazine's letters.
387	★	First Unique Character in a String		Find the first non-repeating character in a string. Return its index or -1.
388	★	Longest Absolute File Path		Find the length of the longest absolute file path in a file system string.
392	★	Is Subsequence		Check if string s is a subsequence of string t.
408	★	Valid Word Abbreviation		Check if an abbreviation matches a word (digits represent skipped characters).
409	★	Longest Palindrome		Find the length of the longest palindrome that can be built from the given letters.
415	★	Add Strings		Add two non-negative integers given as strings. Return the sum as a string.
422	★	Valid Word Square		Given a sequence of words, check whether they form a valid word square (kth row equals kth column).
423	★	Reconstruct Original Digits from English		Given a scrambled string of digit-word letters, decode original digits in order.
434	★	Number of Segments in a String		Count the number of segments (maximal runs of non-space chars) in a string.
443	★	String Compression		Compress an array of chars in-place: 'aabcc' 'a2bc2'. Return new length.
459	★	Repeated Substring Pattern		Check if a string can be built by repeating one of its substrings multiple times.
468	★	Validate IP Address		Validate whether a string is a valid IPv4 or IPv6 address.
500	★	Keyboard Row		Find all words that can be typed on a single row of a QWERTY keyboard.
520	★	Detect Capital		Check if a word is capitalized correctly (all caps, all lower, or first letter only).
524	★	Longest Word in Dictionary through Deleting		Find the longest word in a dictionary that is a subsequence of string s.
541	★	Reverse String II		Reverse the first k characters of every 2k-character block in a string.
551	★	Student Attendance Record I		Check if a student's attendance record is eligible for an award (no late > 1, no absence 3 consecutive).
556	★	Next Greater Element III		Find the next greater element by rearranging the digits of n. Return -1 if not possible or overflow.
557	★	Reverse Words in a String III		Reverse individual words in a string while preserving word order.
592	★	Fraction Addition and Subtraction		Add or subtract fractions, returning the result in lowest terms.
616	★	Add Bold Tag in String		Wrap every substring of s that matches a word in the dictionary with ..., merging overlaps.
657	★	Robot Return to Origin		Check if a robot following instruction string (UDLR) returns to origin.
670	★	Maximum Swap		Swap two digits in a number to get the maximum possible value.
680	★	Valid Palindrome II		Check if a string is a palindrome after deleting at most one character.
681	★	Next Closest Time		Find the next closest time using only the digits from a given time string.
686	★	Repeated String Match		Find the minimum number of times string a must repeat so b is a substring.

#	★	Name	Complexity	Description
709	★	To Lower Case		Convert a string to lowercase.
726	★	Number of Atoms		Parse a chemical formula string and return atom counts in sorted order.
748	★	Shortest Completing Word		Find the shortest word in a list that contains all letters of a license plate (case-insensitive, with multiplicity).
788	★	Rotated Digits		Count numbers in `[1, n]` that become a different valid number when each digit is rotated 180 degrees.
791	★	Custom Sort String		Sort string s so its characters appear in the order given by string order.
794	★	Valid Tic-Tac-Toe State		Decide whether the given tic-tac-toe board is reachable from valid play.
796	★	Rotate String		Check if string t is a rotation of string s.
809	★	Expressive Words		Determine if a query word matches a stretched version of a word in a list.
824	★	Goat Latin		Convert words to Goat Latin: append 'ma', move leading consonants to end + 'ma'.
833	★	Find And Replace in String		Apply non-overlapping indexed replacements: at each index, if `sources[k]` matches, replace it with `targets[k]`.
859	★	Buddy Strings		Check if two strings are 'buddy strings': swapping exactly one pair makes them equal.
953	★	Verifying an Alien Dictionary		Check if words are sorted in a given alien language's alphabetical order.
1055	★	Shortest Way to Form String		Find the minimum number of subsequences of `source` whose concatenation equals `target`. Return -1 if impossible.
1108	★	Defanging an IP Address		Defang an IP address by replacing '.' with '[.]'.
1221	★	Split a String in Balanced Strings		Find the maximum number of balanced strings ('R' count == 'L' count) to split into.
1328	★	Break a Palindrome		Break a palindrome by changing one character to get the lexicographically smallest non-palindrome.
1392	★	Longest Happy Prefix		Find the longest happy prefix (prefix that is also a suffix).
1446	★	Consecutive Characters		Find the maximum number of consecutive same characters in a string.
1554	★	Strings Differ by One Character		Check if any two strings in a list differ by exactly one character.
1556	★	Thousand Separator		Format an integer with a dot as the thousands separator.
1736	★	Latest Time by Replacing Hidden Digits		Find the latest valid time by replacing '?' with appropriate digits.
1816	★	Truncate Sentence		Truncate a sentence to the first k words.
1844	★	Replace All Digits with Characters		Replace each digit at odd positions with the corresponding letter.

Stack / Queue / Heap (55 problems, 55 templated)

#	★	Name	Complexity	Description
20	★	Valid Parentheses	O(n) / O(n)	Determine if a string of brackets '()[]\' is valid --- every open bracket is closed in the correct order.
23	★	Merge k Sorted Lists	O(n log k) / O(k)	Merge k sorted linked lists into one sorted linked list.
32	★	Longest Valid Parentheses		Find the length of the longest valid (well-formed) parentheses substring.
71	★	Simplify Path		Simplify an absolute Unix file path (handle '.', '..', multiple slashes).
84	★	Largest Rectangle in Histogram	O(n) / O(n)	Find the area of the largest rectangle that can be formed in a histogram.
85	★	Maximal Rectangle	O(mn) / O(n)	Find the largest rectangle containing only 1s in a binary matrix.
150	★	Evaluate Reverse Polish Notation		Evaluate a Reverse Polish Notation expression with +, -, *, /.
155	★	Min Stack		Design a stack that supports push, pop, top, and retrieving the minimum in O(1).
215	★	Kth Largest Element in an Array	O(n) avg / O(1)	Find the kth largest element in an unsorted array (not necessarily distinct).
218	★	The Skyline Problem		Given building rectangles, return the skyline as a list of key points.
224	★	Basic Calculator	O(n) / O(n)	Evaluate a string expression with '+', '-', and nested parentheses.
225	★	Implement Stack using Queues		Implement a LIFO stack using only queue operations.
227	★	Basic Calculator II	O(n) / O(n)	Evaluate a string expression with '+', '-', '*', '/' (no parentheses, integer division).
232	★	Implement Queue using Stacks		Implement a queue using two stacks.
295	★	Find Median from Data Stream	O(log n) / O(n)	Design a data structure to find the median of a stream of integers efficiently.
316	★	Remove Duplicate Letters		Remove duplicate letters so each appears once, result is lexicographically smallest.
347	★	Top K Frequent Elements	O(n) / O(n)	Find the k most frequent elements in an integer array.
373	★	Find K Pairs with Smallest Sums	O(k log k) / O(k)	Find the k pairs (u,v) with the smallest sums, one from each of two sorted arrays.
394	★	Decode String		Decode a string encoded as k[encoded_string] where brackets can be nested.
402	★	Remove K Digits		Remove k digits from a number string to get the smallest possible number.
451	★	Sort Characters By Frequency		Sort characters in a string by frequency of occurrence (descending).
456	★	132 Pattern		Check if a 1-3-2 pattern exists in an integer array.
480	★	Sliding Window Median	O(n log k) / O(k)	Find the median in each sliding window of size k across an array.
496	★	Next Greater Element I	O(n) / O(n)	For each element in nums1, find the first greater element to its right in nums2.
502	★	IPO		Maximize profit by choosing at most k projects, each requiring capital and providing profit.
503	★	Next Greater Element II	O(n) / O(n)	Find the next greater element for each element in a circular array.
506	★	Relative Ranks		Assign ranks ("Gold Medal", "Silver Medal", "Bronze Medal", then 4, 5, ...) to athletes by score.
636	★	Exclusive Time of Functions		Given a log of function start/end times, compute exclusive time per function.

#	★	Name	Complexity	Description
678	★	Valid Parenthesis String		Check if a string with '(', ')', '*' (wildcard) can be a valid parenthesis string.
692	★	Top K Frequent Words	O(n log k) / O(n)	Find the k most frequent words (sorted by frequency, then lex).
703	★	Kth Largest Element in a Stream		Design a class to find the kth largest element in a stream.
716	★	Max Stack		Design a stack with push, pop, top, getMax, and popMax operations.
735	★	Asteroid Collision		Simulate asteroids moving in a row: right-moving collide with left-moving.
739	★	Daily Temperatures	O(n) / O(n)	Given temperatures, find the number of days until a warmer temperature.
759	★	Employee Free Time		Find the free time intervals common to all employees given their schedules.
767	★	Reorganize String	O(n log n) / O(n)	Rearrange a string so no two adjacent characters are the same.
772	★	Basic Calculator III	O(n) / O(n)	Evaluate an arithmetic expression with `+`, `-`, `*`, `/`, and parentheses.
844	★	Backspace String Compare		Check if two strings are equal after processing '#' as backspace.
856	★	Score of Parentheses		Calculate the score of a valid parentheses string (empty=0, AB=A+B, (A)=2A or 1 if empty).
862	★	Shortest Subarray with Sum at Least K		Find the length of the shortest subarray with sum k (k can be negative).
907	★	Sum of Subarray Minimums		Find the sum of subarray minimums for all subarrays.
921	★	Minimum Add to Make Parentheses Valid		Find the minimum additions to make a parentheses string valid.
946	★	Validate Stack Sequences		Check if a sequence can be produced by a series of push-pop operations on a stack.
973	★	K Closest Points to Origin	O(n) avg / O(k)	Find the k points closest to the origin.
1047	★	Remove All Adjacent Duplicates In String		Remove all adjacent duplicate characters until no more adjacent duplicates exist.
1086	★	High Five		For each student, compute the average of their top five scores.
1190	★	Reverse Substrings Between Each Pair of Parentheses		Reverse the substrings between each pair of parentheses (innermost first).
1209	★	Remove All Adjacent Duplicates in String II		Remove all adjacent duplicates of length k in a string repeatedly.
1249	★	Minimum Remove to Make Valid Parentheses		Remove the minimum number of parentheses to make the string valid.
1337	★	The K Weakest Rows in a Matrix		Find the k weakest rows in a matrix (fewest soldiers, ties broken by row index).
1541	★	Minimum Insertions to Balance a Parentheses String		Find the minimum insertions to make a string balanced (every ')' needs two '(').
1614	★	Maximum Nesting Depth of the Parentheses		Find the maximum nesting depth of parentheses in a string.
1792	★	Maximum Average Pass Ratio		Maximize the average pass ratio by adding extra students optimally.
1944	★	Number of Visible People in a Queue		Count the number of people each person can see in a queue (taller blocks view).
1985	★	Find the Kth Largest Integer in the Array		Find the kth largest integer in an array of number strings.

DFS / Backtracking (140 problems, 140 templated)

#	★	Name	Complexity	Description
17	★	Letter Combinations of a Phone Number		Given a string of digits 2-9, return all possible letter combinations a phone keypad could represent.
22	★	Generate Parentheses		Generate all combinations of n pairs of well-formed parentheses.
36	★	Valid Sudoku		Determine if a 9×9 Sudoku board is valid (rows, columns, and 3×3 boxes each contain 1-9 with no repeats).
37	★	Sudoku Solver	O(9^m) / O(1)	Solve a Sudoku puzzle by filling empty cells with digits 1-9 such that each row, column, and 3×3 box is valid.
39	★	Combination Sum	O(2^n) / O(n)	Find all unique combinations of candidates (may reuse) that sum to a target.
40	★	Combination Sum II	O(2^n) / O(n)	Find all unique combinations of candidates (each used once) that sum to a target. Candidates may have duplicates.
46	★	Permutations	O(n!) / O(n)	Return all possible permutations of a distinct-integer array.
47	★	Permutations II	O(n!) / O(n)	Return all unique permutations of an array that may contain duplicates.
51	★	N-Queens	O(n!) / O(n)	Place n queens on an n×n chessboard so no two attack each other. Return all distinct solutions.
60	★	Permutation Sequence	O(n) / O(n)	Find the kth permutation sequence of numbers 1 to n.
77	★	Combinations		Return all combinations of k numbers from 1 to n.
78	★	Subsets	O(2^n) / O(n)	Return all possible subsets (power set) of a distinct-integer array.
79	★	Word Search	O(mn4^L) / O(L)	Search for a word in an m×n grid of characters by adjacent (up/down/left/right) traversal without reuse.
90	★	Subsets II	O(2^n) / O(n)	Return all unique subsets of an array that may contain duplicates.
93	★	Restore IP Addresses		Restore all possible valid IP address strings from a string of digits.
94	★	Binary Tree Inorder Traversal		Return the inorder traversal (left--root--right) of a binary tree.
95	★	Unique Binary Search Trees II		Generate all structurally unique BSTs with values 1..n.
98	★	Validate Binary Search Tree	O(n) / O(h)	Validate whether a binary tree is a valid Binary Search Tree.
99	★	Recover Binary Search Tree		Recover a BST where exactly two nodes have been swapped.
100	★	Same Tree		Check if two binary trees are structurally identical with the same node values.
101	★	Symmetric Tree		Check if a binary tree is symmetric (mirror image of itself).
102	★	Binary Tree Level Order Traversal	O(n) / O(n)	Return node values level by level (left to right) in a binary tree.
103	★	Binary Tree Zigzag Level Order Traversal	O(n) / O(n)	Return node values in zigzag level order (left-to-right for odd levels, right-to-left for even).
104	★	Maximum Depth of Binary Tree	O(n) / O(h)	Find the maximum depth (number of nodes along the longest root-to-leaf path) of a binary tree.
105	★	Construct Binary Tree from Preorder and Inorder Traversal		Reconstruct a binary tree from its preorder and inorder traversal arrays.
106	★	Construct Binary Tree from Inorder and Postorder Traversal		Reconstruct a binary tree from its inorder and postorder traversal arrays.
107	★	Binary Tree Level Order Traversal II		Return node values level by level from bottom to top.
108	★	Convert Sorted Array to Binary Search Tree		Convert a sorted integer array to a height-balanced BST.
110	★	Balanced Binary Tree	O(n) / O(h)	Determine if a binary tree is height-balanced (every node's subtrees differ in depth by at most 1).
111	★	Minimum Depth of Binary Tree	O(n) / O(h)	Find the minimum depth of a binary tree (shortest path from root to a leaf).

#	★	Name	Complexity	Description
112	★	Path Sum		Check if a root-to-leaf path exists whose node values sum to a given target.
113	★	Path Sum II		Find all root-to-leaf paths whose values sum to a given target.
114	★	Flatten Binary Tree to Linked List		Flatten a binary tree into a linked list in-place (preorder, using right pointers).
116	★	Populating Next Right Pointers in Each Node		Populate each node's 'next' pointer to its right neighbor (perfect binary tree). Use O(1) extra space.
117	★	Populating Next Right Pointers in Each Node II		Same as 116 but for an arbitrary binary tree.
124	★	Binary Tree Maximum Path Sum	O(n) / O(h)	Find the maximum path sum in a binary tree (path can start/end at any node).
129	★	Sum Root to Leaf Numbers		Sum all root-to-leaf numbers where each path forms a number (root digit is most significant).
144	★	Binary Tree Preorder Traversal		Return the preorder traversal (root → left → right) of a binary tree.
145	★	Binary Tree Postorder Traversal		Return the postorder traversal (left → right → root) of a binary tree.
199	★	Binary Tree Right Side View	O(n) / O(n)	Return the values visible from the right side of a binary tree (one per level).
216	★	Combination Sum III	O(C(9,k)) / O(k)	Find all combinations of k distinct numbers from 1--9 that sum to n.
222	★	Count Complete Tree Nodes		Count nodes in a complete binary tree without visiting all nodes. Faster than O(n).
226	★	Invert Binary Tree		Invert (mirror) a binary tree.
230	★	Kth Smallest Element in a BST	O(n) / O(h)	Find the kth smallest element in a BST.
235	★	Lowest Common Ancestor of a Binary Search Tree		Find the lowest common ancestor of two nodes in a BST.
236	★	Lowest Common Ancestor of a Binary Tree		Find the lowest common ancestor of two nodes in a binary tree (not necessarily a BST).
241	★	Different Ways to Add Parentheses		Given an expression string, compute all possible results from different ways to parenthesize it.
250	★	Count Univalue Subtrees		Count subtrees in which all nodes share the same value.
255	★	Verify Preorder Sequence in Binary Search Tree		Verify whether an array of integers is the correct preorder traversal of a BST.
257	★	Binary Tree Paths		Return all root-to-leaf paths in a binary tree as strings ('1->2->5').
267	★	Palindrome Permutation II		Return all palindrome permutations of a string.
270	★	Closest Binary Search Tree Value	O(h) / O(1)	Find the value in a BST closest to a given target.
272	★	Closest Binary Search Tree Value II		Find k values in a BST closest to target.
282	★	Expression Add Operators		Insert operators +, -, * between digits of a string num to reach a target value. Return all expressions.
285	★	Inorder Successor in BST	O(h) / O(1)	Find the inorder successor of a given node in a BST.
291	★	Word Pattern II		Check if a string s follows a given pattern where pattern chars map to non-overlapping substrings.
293	★	Flip Game		Given a string of '+' and '-', return all states after flipping one "++" to "--".
294	★	Flip Game II		Determine if the first player can guarantee a win in Flip Game.
297	★	Serialize and Deserialize Binary Tree		Serialize and deserialize a binary tree to/from a string.
298	★	Binary Tree Longest Consecutive Sequence		Find the longest consecutive sequence in a binary tree (parent to child, values increasing by 1).
301	★	Remove Invalid Parentheses		Remove the minimum number of parentheses to make the string valid. Return all results.

#	★	Name	Complexity	Description
306	★	Additive Number		Check if a string forms an additive number (each number = sum of two preceding).
314	★	Binary Tree Vertical Order Traversal		Return the binary tree's node values ordered by column, then by row.
320	★	Generalized Abbreviation		Return all possible abbreviations of a word.
339	★	Nested List Weight Sum		Compute the weighted sum of a nested integer list where weight equals depth.
364	★	Nested List Weight Sum II		Sum each integer weighted by its inverse depth (deepest integers have weight 1).
386	★	Lexicographical Numbers		Return integers 1..n in lexicographical order.
404	★	Sum of Left Leaves		Find the sum of all left leaves in a binary tree.
426	★	Convert Binary Search Tree to Sorted Doubly Linked List		Convert a BST to a sorted circular doubly linked list in-place.
429	★	N-ary Tree Level Order Traversal		Return the level order traversal of an N-ary tree.
437	★	Path Sum III	O(n) / O(n)	Count paths in a binary tree that sum to a given value (path goes downward, any node to any node).
440	★	K-th Smallest in Lexicographical Order		Find the kth smallest integer in 1..n by lexicographical order.
449	★	Serialize and Deserialize BST		Serialize and deserialize a BST compactly.
450	★	Delete Node in a BST		Delete a given key from a BST and return the updated root.
473	★	Matchsticks to Square		Determine if n matchsticks can form exactly 4 equal-length sides of a square.
488	★	Zuma Game		Find the minimum number of marbles to insert in Zuma to clear the board.
489	★	Robot Room Cleaner		Design a robot cleaner: given room-cleaning robot with no map, clean the entire room.
491	★	Increasing Subsequences		Return all increasing subsequences of length ≥ 2 (input may have duplicates).
501	★	Find Mode in Binary Search Tree		Find the mode(s) in a BST (values that appear most frequently).
508	★	Most Frequent Subtree Sum		Find the subtree sum(s) that occur most frequently.
510	★	Inorder Successor in BST II		Find the inorder successor of a node given only a parent pointer (no root).
513	★	#513	O(n) / O(n)	Find leftmost value in last row of binary tree.
515	★	Find Largest Value in Each Tree Row		Return the maximum value found at each level.
526	★	Beautiful Arrangement		Count 'beautiful arrangements': n numbers placed so each divides or is divisible by its position.
536	★	Construct Binary Tree from String		Construct a binary tree from its string representation '4(2(3)(1))(6(5))'.
538	★	Convert BST to Greater Tree		Convert each BST node's value to the sum of all values greater than or equal to it.
543	★	Diameter of Binary Tree	O(n) / O(h)	Find the diameter of a binary tree (longest path between any two nodes).
545	★	Boundary of Binary Tree		Return the boundary of a binary tree: left boundary + leaves + right boundary (no duplicates).
549	★	Binary Tree Longest Consecutive Sequence II	O(n) / O(h)	Find the longest consecutive sequence in a binary tree (can go up or down, must be consecutive).
559	★	Maximum Depth of N-ary Tree		Find the maximum depth of an N-ary tree.
563	★	Binary Tree Tilt		Find the total tilt of a binary tree (sum of leftSum - rightSum for all nodes).
572	★	Subtree of Another Tree		Check if tree t is a subtree of tree s.
589	★	N-ary Tree Preorder Traversal		Return the preorder traversal of an N-ary tree.
590	★	N-ary Tree Postorder Traversal		Return the postorder traversal of an N-ary tree.

#	★	Name	Complexity	Description
606	★	Construct String from Binary Tree		Construct a string from a binary tree using preorder with parentheses.
617	★	Merge Two Binary Trees		Merge two binary trees by summing overlapping nodes.
637	★	Average of Levels in Binary Tree		Return the average value of each level of a binary tree.
652	★	Find Duplicate Subtrees		Find all duplicate subtrees in a binary tree; return root of each.
653	★	Two Sum IV - Input is a BST		Find two numbers in a BST that sum to a given target.
654	★	Maximum Binary Tree		Build the maximum binary tree: root is array max, left from left subarray, right from right.
662	★	Maximum Width of Binary Tree	O(n) / O(n)	Return the maximum width of any level of a binary tree. Width is defined as the length between the leftmost and rightmost non-null nodes, including all the null nodes in between.
671	★	Second Minimum Node In a Binary Tree		In a tree where each node's value is the min of its children, find the second smallest value overall.
679	★	24 Game		Check if 4 numbers with any order and any of +,-,*,/ can produce 24.
687	★	#687	O(n) / O(h)	Longest univalue path --- edges between nodes of equal value.
698	★	Partition to K Equal Sum Subsets		Determine if an array can be split into k subsets of equal sum.
700	★	Search in a Binary Search Tree		Search for a value in a BST.
701	★	Insert into a Binary Search Tree		Insert a value into a BST and return the updated root.
742	★	Closest Leaf in a Binary Tree		Find the node in a binary tree closest to the target that is a leaf.
776	★	Split BST		Split a BST into two trees: one with values V and one with values > V.
784	★	Letter Case Permutation		Return all strings formed by toggling the case of letters in a string.
814	★	Binary Tree Pruning		Prune a binary tree by removing subtrees containing no 1s.
842	★	Split Array into Fibonacci Sequence		Check if a string can be split into a Fibonacci-like sequence.
863	★	All Nodes Distance K in Binary Tree		Find all nodes at distance k from a target node in a binary tree.
865	★	Smallest Subtree with all the Deepest Nodes		Find the smallest subtree containing all the deepest leaves.
889	★	Construct Binary Tree from Preorder and Postorder Traversal		Reconstruct a binary tree from preorder and postorder traversals (any valid tree).
897	★	Increasing Order Search Tree		Transform a BST into an increasing-order search tree (right-skewed).
919	★	Complete Binary Tree Inserter		Design a complete binary tree inserter with O(1) insert.
932	★	Beautiful Array		Construct a beautiful array where no three elements form an arithmetic progression.
938	★	Range Sum of BST		Return the sum of all BST node values within the range [low, high].
951	★	Flip Equivalent Binary Trees		Check if two binary trees are flip-equivalent (can make them identical by flipping children).
958	★	Check Completeness of a Binary Tree		Check if a binary tree is a complete binary tree.
987	★	Vertical Order Traversal of a Binary Tree		Return node values grouped by vertical position (column), then by row, then by value.
988	★	Smallest String Starting From Leaf		Find the lexicographically smallest string from root-to-leaf path.
993	★	Cousins in Binary Tree		Check if two nodes in a binary tree are cousins (same depth, different parents).
998	★	Maximum Binary Tree II		Insert a value into a maximum tree built by appending the value to the original array's end.

#	★	Name	Complexity	Description
1008	★	Construct Binary Search Tree from Preorder Traversal		Construct a BST from its preorder traversal.
1026	★	Maximum Difference Between Node and Ancestor		Find the maximum difference between a node and its ancestor in a binary tree.
1038	★	Binary Search Tree to Greater Sum Tree		Convert each BST node to the sum of all nodes with a greater or equal value.
1087	★	Brace Expansion		Return all strings formed by expanding brace expressions (comma-separated choices).
1104	★	Path In Zigzag Labelled Binary Tree		Find the parent node in a zigzag-labeled infinite binary tree.
1110	★	Delete Nodes And Return Forest		Delete given nodes from a binary tree and return the roots of the remaining forest.
1123	★	Lowest Common Ancestor of Deepest Leaves		Find the LCA of the deepest leaves in a binary tree.
1161	★	Maximum Level Sum of a Binary Tree		Find the level of a binary tree with the maximum sum of node values.
1214	★	Two Sum BSTs		Given two BSTs and a target, decide if a value from each sums to the target.
1305	★	All Elements in Two Binary Search Trees		Return all elements from two BSTs in sorted order.
1361	★	Validate Binary Tree Nodes		Given child arrays, verify the nodes form exactly one valid binary tree.
1382	★	Balance a Binary Search Tree		Convert a BST to a balanced BST.
1522	★	Diameter of N-Ary Tree		Find the diameter of an N-ary tree.
1644	★	Lowest Common Ancestor of a Binary Tree II		Find the LCA of two nodes in a binary tree where nodes may not exist.
1650	★	Lowest Common Ancestor of a Binary Tree III		Find the LCA of two nodes given parent pointers (no root access).

Graph (58 problems, 58 templated)

#	★	Name	Complexity	Description
126	★	Word Ladder II		Find all shortest transformation sequences from beginWord to endWord, changing one letter at a time.
127	★	Word Ladder	O(V+E) / O(V)	Find the length of the shortest transformation sequence from beginWord to endWord (one letter at a time, each word must be in wordList).
130	★	Surrounded Regions	O(mn) / O(mn)	Capture all 'O' regions not connected to the border (surrounded by 'X'); flip them to 'X'.
133	★	Clone Graph		Deep-copy (clone) an undirected graph where each node has a value and a neighbors list.
200	★	Number of Islands	O(mn) / O(mn)	Count the number of islands (connected regions of '1's) in a 2D binary grid.
207	★	Course Schedule	O(V+E) / O(V+E)	Determine if all n courses can be finished given prerequisite pairs (detect cycle in directed graph).
210	★	Course Schedule II	O(V+E) / O(V+E)	Return a valid course order given prerequisites (topological sort). Return [] if impossible.
261	★	Graph Valid Tree	O((V+E)) / O(V)	Determine if n nodes and given edges form a valid undirected tree.
269	★	Alien Dictionary	O(V+E) / O(V+E)	Given a sorted list of alien words, determine the order of characters in the alien alphabet. Return the order as a string, or empty string if invalid.
277	★	Find the Celebrity		Find the celebrity in a group of n people: everyone knows them, they know nobody.
286	★	Walls and Gates	O(mn) / O(mn)	Walls and Gates: fill each empty room (INF) with its distance to the nearest gate (0).
296	★	Best Meeting Point		Find the point minimizing total Manhattan distance to a set of buildings on a grid.
305	★	Number of Islands II		Count islands after each cell is added (online algorithm).
310	★	Minimum Height Trees	O(V+E) / O(V+E)	Find all minimum height trees in an undirected tree graph.
317	★	Shortest Distance from All Buildings		Find the empty land cell minimizing total walking distance to all buildings.
323	★	Number of Connected Components in an Undirected Graph	O((V+E)) / O(V)	Count connected components in an undirected graph.
329	★	Longest Increasing Path in a Matrix		Find the length of the longest increasing path in a matrix (4-directional movement).
399	★	Evaluate Division		Given equations and ratios, evaluate new queries (transitive division).
407	★	Trapping Rain Water II		Given a 2D height map, compute how much water it can trap after raining.
417	★	Pacific Atlantic Water Flow	O(mn) / O(mn)	Find all cells from which water can flow to both Pacific and Atlantic oceans (4-directional).
463	★	Island Perimeter		Calculate the perimeter of islands in a grid.
529	★	Minesweeper		Simulate the first click in Minesweeper: reveal cells per the game rules.
542	★	01 Matrix		For each cell in a binary matrix, find the distance to the nearest 0.
547	★	Number of Provinces		Count the number of provinces (connected components) among n cities given friendship pairs.
684	★	#684	O(E) / O(V)	Remove one redundant edge to make graph a tree.
694	★	Number of Distinct Islands		Count the number of distinct islands in a binary grid (by shape, not position).
695	★	Max Area of Island	O(mn) / O(mn)	Find the maximum area of any island in a binary grid.
721	★	#721	O(n^2) / O(n)	Merge email accounts sharing a common email address.

#	★	Name	Complexity	Description
733	★	Flood Fill		Flood-fill a pixel in an image: change all connected same-color pixels.
743	★	Network Delay Time	O((V+E) log V) / O(V+E)	Find the minimum time for a signal to reach all nodes in a network (Dijkstra).
752	★	Open the Lock		Find the minimum number of turns to open a 4-digit lock from '0000', avoiding dead-ends.
778	★	Swim in Rising Water		Find the minimum time to swim from top-left to bottom-right of a grid (elevation = wait time).
785	★	Is Graph Bipartite?		Check if an undirected graph can be 2-colored (bipartite).
797	★	All Paths From Source to Target		Find all paths from node 0 to node n-1 in a DAG.
802	★	Find Eventual Safe States		Find all safe nodes (nodes from which all paths lead to terminal nodes) in a DAG.
815	★	Bus Routes		Find the minimum number of buses to take to travel from source to target stop.
827	★	Making A Large Island	O(mn) / O(mn)	Connect two islands to maximize island size; return the maximum.
847	★	Shortest Path Visiting All Nodes		Find the shortest path visiting all nodes in an undirected graph.
851	★	Loud and Rich		Find the quietest person among those richer than each person.
886	★	Possible Bipartition		Check if people can be split into two groups with no one in the same group disliking each other.
909	★	Snakes and Ladders	O(n^2) / O(n^2)	Find the minimum number of dice rolls to reach the last square (snakes and ladders).
913	★	Cat and Mouse		A mouse (start node 1) and cat (start node 2) move on a graph; mouse wins by reaching node 0, cat wins by catching the mouse, else draw. Return 0/1/2 with optimal play.
934	★	Shortest Bridge		Find the shortest bridge between two islands in a binary matrix.
994	★	Rotting Oranges	O(mn) / O(mn)	Find the minimum time to rot all fresh oranges (multi-source BFS).
1034	★	Coloring A Border		Color cells on the border of an island (connected component) with a given color.
1091	★	Shortest Path in Binary Matrix	O(mn) / O(mn)	Find the shortest path from top-left to bottom-right in a binary matrix (8-directional).
1102	★	Path With Maximum Minimum Value		Find the path from top-left to bottom-right maximizing the minimum value on the path.
1136	★	Parallel Courses		Find the minimum number of months to complete all courses given prerequisites and durations.
1162	★	As Far from Land as Possible		In a grid of `0` (water) and `1` (land), find the water cell whose distance to the nearest land is maximized; return that distance, or -1.
1197	★	Minimum Knight Moves		Find the minimum knight moves from origin to a target on an infinite board.
1202	★	Smallest String With Swaps		Swap characters at given index pairs to get the lexicographically smallest string.
1245	★	Tree Diameter		Find the diameter of an N-ary tree (longest path between any two nodes).
1293	★	Shortest Path in a Grid with Obstacles Elimination		Find the shortest path from start to end in a grid, eliminating at most k obstacles.
1319	★	Number of Operations to Make Network Connected		Find the minimum number of additional connections to make a network fully connected.
1514	★	#1514	O(E log V) / O(V+E)	Max probability path between two nodes.
1559	★	Detect Cycles in 2D Grid		Detect if a 2D grid contains a cycle of the same character.
1631	★	Path With Minimum Effort	O(mn log mn) / O(mn)	Find the path from top-left to bottom-right minimizing the maximum absolute difference.

#	★	Name	Complexity	Description
1743	★	Restore the Array From Adjacent Pairs		Restore the original array from a list of all adjacent pairs.

Greedy (28 problems, 28 templated)

#	★	Name	Complexity	Description
45	★	Jump Game II	O(n) / O(1)	Find the minimum number of jumps to reach the last index. Each element is max jump length.
55	★	Jump Game	O(n) / O(1)	Given array of jump lengths, determine if you can reach the last index from index 0.
56	★	Merge Intervals	O(n log n) / O(n)	Merge all overlapping intervals in a collection and return non-overlapping sorted intervals.
57	★	Insert Interval	O(n) / O(n)	Insert a new interval into a sorted non-overlapping list of intervals, merging if necessary.
134	★	Gas Station	O(n) / O(1)	Find the starting gas station index from which you can complete a circular route. Return -1 if impossible.
135	★	Candy		Assign minimum candies to children in a line: each child 1 candy; child with higher rating than neighbor gets more.
252	★	Meeting Rooms	O(n log n) / O(1)	Determine if a person can attend all meetings (no overlapping intervals).
253	★	Meeting Rooms II	O(n log n) / O(n)	Find the minimum number of conference rooms required to hold all meetings.
274	★	H-Index		Given citation counts, find the largest h such that h papers have citations each.
358	★	Rearrange String k Distance Apart	O(n log k) / O(k)	Rearrange a string so that the same characters are at least `k` distance apart. Return `""` if impossible.
406	★	Queue Reconstruction by Height		Reconstruct a queue of people given (height, k) pairs where k = taller-or-equal people in front.
435	★	Non-overlapping Intervals	O(n log n) / O(1)	Find the minimum number of intervals to remove so the rest don't overlap.
452	★	Minimum Number of Arrows to Burst Balloons		Find the minimum number of arrows to burst all balloons (each arrow goes through all balloons at that x).
455	★	Assign Cookies		Assign cookies greedily: each child needs minimum size, each cookie satisfies one child.
575	★	Distribute Candies		Distribute n candies (n/2 to eat) maximizing the number of distinct types.
605	★	Can Place Flowers		Determine if n flowers can be planted in a flowerbed without adjacent flowers.
621	★	Task Scheduler	O(n) / O(1)	Find the minimum intervals needed to execute n tasks with cooldown k between same-type tasks.
630	★	Course Schedule III		Find the maximum courses you can take given (duration, deadline) pairs.
646	★	Maximum Length of Pair Chain		Find the longest chain of pairs [a,b] where b < next_a.
763	★	Partition Labels		Partition a string into as many parts as possible so each letter appears in at most one part.
881	★	Boats to Save People		Find the minimum number of boats to rescue people given weight limit 2 per boat.
986	★	Interval List Intersections		Find all intersections of two lists of intervals.
1005	★	Maximize Sum Of Array After K Negations		Given an array and `k`, perform exactly `k` operations, each negating one element (the same index may be chosen repeatedly), to maximize the array sum.
1094	★	Car Pooling		Determine if a car pool trip is feasible given passenger pick-up and drop-off locations.
1272	★	Remove Interval		Given a sorted list of disjoint intervals and an interval `toBeRemoved`, return the set of intervals after removing every point that lies inside `toBeRemoved`.
1326	★	Minimum Number of Taps to Open to Water a Garden		Find the minimum number of taps to open to water the entire garden.

#	★	Name	Complexity	Description
1353	★	Maximum Number of Events That Can Be Attended		Find the maximum number of events you can attend (one per day).
1488	★	Avoid Flood in The City		Schedule drying days to prevent flooding given a list of rains.

Dynamic Programming (86 problems, 86 templated)

#	★	Name	Complexity	Description
5	★	Longest Palindromic Substring	O(n^2) / O(1)	Find the longest palindromic substring within a given string.
10	★	Regular Expression Matching	O(mn) / O(mn)	Implement regular expression matching with '.' (any single char) and '*' (zero or more of preceding). Must match entire string.
44	★	Wildcard Matching		Match a string against a pattern with '?' (any single char) and '*' (any sequence including empty).
53	★	Maximum Subarray		Find the contiguous subarray with the largest sum (at least one element).
62	★	Unique Paths	O(mn) / O(n)	Count distinct paths from the top-left to the bottom-right of an m×n grid (only right or down moves).
63	★	Unique Paths II	O(mn) / O(n)	Count distinct paths in an m×n grid with obstacles (0=open, 1=obstacle).
64	★	Minimum Path Sum	O(mn) / O(1)	Find the path from top-left to bottom-right of a grid with minimum sum.
70	★	Climbing Stairs	O(n) / O(1)	Count the number of ways to climb n stairs, taking 1 or 2 steps at a time.
72	★	Edit Distance	O(mn) / O(mn)	Find the minimum number of edit operations (insert/delete/replace) to convert word1 to word2.
87	★	#87	O(n^3) / O(n^2)	Is s2 a scramble of s1 (recursive splits)?
91	★	Decode Ways	O(n) / O(1)	Count the number of ways to decode a string of digits into letters (A=1 … Z=26).
96	★	Unique Binary Search Trees	O(n^2) / O(n)	Count the number of structurally unique BSTs with exactly n nodes.
97	★	Interleaving String		Determine if s3 is formed by interleaving s1 and s2.
115	★	Distinct Subsequences	O(mn) / O(mn)	Count distinct subsequences of s that equal t.
120	★	Triangle		Find the minimum path sum from top to bottom of a triangle, moving to adjacent numbers on the row below.
121	★	Best Time to Buy and Sell Stock	O(n) / O(1)	Given daily stock prices, find the maximum profit from one buy-sell transaction (buy before sell).
122	★	Best Time to Buy and Sell Stock II	O(n) / O(1)	Maximize profit by completing as many buy-sell transactions as desired (no simultaneous holdings).
123	★	Best Time to Buy and Sell Stock III	O(n) / O(1)	Maximize profit with at most two buy-sell transactions (no simultaneous holdings).
131	★	Palindrome Partitioning	O(n2^n) / O(n)	Partition a string into all possible subsets where every substring is a palindrome.
132	★	Palindrome Partitioning II		Find the minimum number of cuts to partition a string so every part is a palindrome.
139	★	Word Break	O(n^2) / O(n)	Determine if a string can be segmented into space-separated words from a dictionary.
140	★	Word Break II		Return all ways to segment a string into space-separated words from a dictionary.
152	★	Maximum Product Subarray		Find the contiguous subarray with the largest product.
174	★	Dungeon Game		Find the minimum initial health to traverse a dungeon grid from top-left to bottom-right.
188	★	Best Time to Buy and Sell Stock IV		Maximize stock profit with at most k buy-sell transactions.
198	★	House Robber	O(n) / O(1)	Maximize the amount stolen from houses in a row; cannot rob two adjacent houses.
213	★	House Robber II	O(n) / O(1)	Maximize robbery from a circular row of houses (first and last are adjacent).

#	★	Name	Complexity	Description
221	★	Maximal Square	O(mn) / O(mn)	Find the largest square containing only 1s in a binary matrix. Return its area.
264	★	Ugly Number II		Find the nth ugly number (positive number whose prime factors are only 2, 3, 5).
279	★	Perfect Squares	O(n√n) / O(n)	Find the minimum number of perfect squares summing to n.
300	★	Longest Increasing Subsequence	O(n^2) / O(n)	Find the length of the longest strictly increasing subsequence.
309	★	Best Time to Buy and Sell Stock with Cooldown	O(n) / O(1)	Find the maximum profit from stock with a 1-day cooldown after selling.
312	★	Burst Balloons	$O(n^3)$ / O(n^2)	Burst all balloons to maximize collected coins. Coins = left * burst * right.
313	★	Super Ugly Number		Find the nth super ugly number (prime factors only from a given list).
322	★	Coin Change	O(amount) / O(amount)	Find the minimum number of coins to make a given amount (coins can be reused).
337	★	House Robber III		Find the maximum amount you can rob from a binary tree of houses (no two adjacent nodes).
343	★	Integer Break		Break n into at least two positive integers to maximize their product.
368	★	Largest Divisible Subset	O(n^2) / O(n)	Find the largest subset where every pair has the larger divisible by the smaller.
375	★	Guess Number Higher or Lower II		Find the minimum cost to guarantee finding a number in range [1,n] using a guessing game.
376	★	Wiggle Subsequence		Find the length of the longest wiggle subsequence (alternating up-down differences).
377	★	Combination Sum IV	O(target) / O(target)	Count the number of possible combinations that add up to a target (order matters).
413	★	Arithmetic Slices		Count the number of arithmetic slices (subarrays of length 3 with equal differences).
416	★	Partition Equal Subset Sum	O(n) / O(sum)	Determine if an integer array can be partitioned into two subsets with equal sum.
446	★	Arithmetic Slices II - Subsequence		Count the number of arithmetic slices that are subsequences of an array.
474	★	#474	O(l) / O(mn)	Max strings from list fitting within m zeros and n ones.
486	★	Predict the Winner		Two players alternately pick from either end of an array; determine if the first player wins.
494	★	Target Sum	O(n) / O(sum)	Count the number of ways to assign + or - to array elements to reach a target sum.
509	★	Fibonacci Number		Compute the nth Fibonacci number.
516	★	Longest Palindromic Subsequence	O(n^2) / O(n^2)	Find the length of the longest palindromic subsequence in a string.
518	★	Coin Change 2	O(amount) / O(amount)	Count the number of combinations of coins that sum to an amount (coins may be reused).
552	★	Student Attendance Record II		Count valid attendance records of length n (at most 1 absence, never 3+ consecutive lates).
576	★	Out of Boundary Paths		Count paths from a given cell that leave an m×n grid in exactly maxMove steps.
583	★	Delete Operation for Two Strings	O(mn) / O(mn)	Find the minimum number of deletions to make two strings equal.
600	★	Non-negative Integers without Consecutive Ones		Count integers in `[0, n]` whose binary representation has no two adjacent 1 bits.
629	★	K Inverse Pairs Array		Count permutations of 1..n with exactly k inverse pairs.
638	★	Shopping Offers		Minimize the cost to purchase items given special-offer bundles.

#	★	Name	Complexity	Description
639	★	Decode Ways II		Count decodings of a digit string where `*` is a wildcard for digits 1-9, modulo 1e9+7.
647	★	Palindromic Substrings	O(n^2) / O(1)	Count all substrings of a string that are palindromes.
650	★	2 Keys Keyboard		Find minimum steps to get exactly n 'A's on a screen using only Copy All and Paste.
673	★	Number of Longest Increasing Subsequence		Find the number of longest increasing subsequences.
674	★	Longest Continuous Increasing Subsequence		Find the length of the longest continuous (contiguous) strictly increasing subarray.
688	★	Knight Probability in Chessboard		Find the probability that a knight stays on an n×n board after exactly k moves.
714	★	Best Time to Buy and Sell Stock with Transaction Fee	O(n) / O(1)	Find the maximum profit from unlimited transactions with a transaction fee per trade.
718	★	Maximum Length of Repeated Subarray		Find the maximum length of a subarray that appears in both arrays.
740	★	Delete and Earn		Delete and earn: delete nums[i] to earn nums[i] points, but must delete all adjacent values.
746	★	Min Cost Climbing Stairs	O(n) / O(1)	Find the minimum cost to climb a staircase (pay to step on, can skip one).
787	★	Cheapest Flights Within K Stops	O(E log E) / O(V+E)	Find the cheapest flight from src to dst with at most k stops.
873	★	Length of Longest Fibonacci Subsequence		Find the length of the longest Fibonacci-like subsequence in a sorted array.
887	★	Super Egg Drop		Find the minimum number of moves to determine the critical floor for egg drops.
918	★	Maximum Sum Circular Subarray		Find the maximum subarray sum where the array is circular (subarray may wrap around).
931	★	#931	O(mn) / O(1)	Min sum of a falling path through an n×n matrix.
935	★	Knight Dialer		Count distinct phone numbers of length n that a knight can dial.
983	★	Minimum Cost For Tickets		Find the minimum cost to cover all travel days using 1-, 7-, or 30-day passes.
1027	★	Longest Arithmetic Subsequence		Find the length of the longest arithmetic subsequence in an array.
1048	★	Longest String Chain		Find the longest string chain where each word is a predecessor of the next.
1049	★	#1049	O(n) / O(sum)	Smallest possible weight of last remaining stone.
1137	★	N-th Tribonacci Number		Return the nth Tribonacci number (each term is the sum of the previous three).
1143	★	Longest Common Subsequence	O(mn) / O(mn)	Find the length of the longest common subsequence of two strings.
1216	★	Valid Palindrome III		Check if a string is a k-palindrome (palindrome after removing at most k chars).
1218	★	Longest Arithmetic Subsequence of Given Difference		Find the longest arithmetic subsequence with a given difference.
1269	★	Number of Ways to Stay in the Same Place After Some Steps		Count the number of ways to stay at position 0 after exactly n steps on an array of length arrLen.
1312	★	#1312	O(n^2) / O(n^2)	Min insertions to make string a palindrome.
1395	★	Count Number of Teams		Count triplets `(i, j, k)` with `i < j < k` whose ratings are strictly increasing or strictly decreasing.

#	★	Name	Complexity	Description
1567	★	Maximum Length of Subarray With Positive Product		Find the maximum length subarray with a positive product.
1696	★	Jump Game VI		Find the maximum score reaching the last index, jumping 1 to k steps.
1884	★	Egg Drop With 2 Eggs and N Floors		Find minimum moves to determine a critical floor using 2 eggs and n floors.

Bit Manipulation (16 problems, 16 templated)

#	★	Name	Complexity	Description
136	★	Single Number	O(n) / O(1)	Find the single element in an array where every other element appears exactly twice.
137	★	Single Number II		Find the single element in an array where every other element appears exactly three times.
190	★	Reverse Bits		Reverse all bits of a 32-bit unsigned integer.
191	★	Number of 1 Bits	O(1) / O(1)	Count the number of '1' bits in a 32-bit integer (Hamming weight).
201	★	#201	O(log n) / O(1)	Bitwise AND of all numbers in range [left, right].
231	★	Power of Two	O(1) / O(1)	Determine if a number is a power of two.
260	★	Single Number III	O(n) / O(1)	Find two non-repeating numbers in an array where all others appear twice.
268	★	Missing Number	O(n) / O(1)	Find the single missing number in an array of n distinct integers in range [0, n].
318	★	Maximum Product of Word Lengths		Find the maximum product of lengths of two words with no common letters.
338	★	Counting Bits	O(n) / O(n)	Return an array counting the number of 1 bits for each integer from 0 to n.
342	★	Power of Four	O(1) / O(1)	Determine if a number is a power of four.
371	★	Sum of Two Integers		Add two integers without using + or - operators.
405	★	Convert a Number to Hexadecimal		Given an integer `num`, return its hexadecimal representation as a lowercase string. For negative numbers use two's complement.
461	★	Hamming Distance		Calculate Hamming distance between two integers (positions where bits differ).
476	★	Number Complement		Find the complement of an integer (flip all bits up to the most significant bit).
957	★	Prison Cells After N Days		Find the prison cell configuration after n days of simultaneous updates.

Design (25 problems, 24 templated)

#	★	Name	Complexity	Description
146	★	LRU Cache	O(1) / O(capacity)	Design an LRU (Least Recently Used) Cache with O(1) get and put operations.
170	★	Two Sum III - Data structure design		Design a Two Sum data structure with add() and find() operations.
173	★	Binary Search Tree Iterator		Implement a Binary Search Tree iterator with hasNext() and next() using O(h) memory.
244	★	Shortest Word Distance II		Design data structure for finding shortest word distance between two words in a list, supporting repeated queries.
245	★	Shortest Word Distance III		Given a word list and two words (which may be equal), return the shortest index distance. When `word1 == word2`, find the closest pair of distinct occurrences of that word.
251	★	Flatten 2D Vector		Design an iterator over a 2D vector with `next()` and `hasNext()` that yields all elements in row-major order.
281	★	Zigzag Iterator		Given two lists, design an iterator that returns elements alternating between them; when one is exhausted, continue with the other.
341	★	Flatten Nested List Iterator		Implement an iterator that flattens a nested list of integers.
346	★	Moving Average from Data Stream		Compute the moving average from a stream of integers given a window size.
348	★	Design Tic-Tac-Toe		Design a Tic-Tac-Toe game that detects a winner in O(1) per move.
380	★	Insert Delete GetRandom O(1)		Design a set with O(1) insert, remove, and random element retrieval.
381	★	Insert Delete GetRandom O(1) - Duplicates allowed		Same as 380 but allows duplicates.
460	★	LFU Cache	O(1) / O(capacity)	Design an LFU (Least Frequently Used) Cache with O(1) get and put.
519	★	Random Flip Matrix		Design a matrix with random flip and reset; each 0-cell can be flipped to 1 exactly once.
622	★	Design Circular Queue		Design a circular queue (ring buffer) with fixed capacity.
641	★	Design Circular Deque		Design a double-ended circular queue with maxSize capacity.
705	★	Design HashSet		Design a HashSet with add, remove, and contains operations.
706	★	Design HashMap		Design a HashMap with put, get, and remove operations.
707	★	Design Linked List		Design a doubly linked list with index-based get, addAtHead, addAtTail, addAtIndex, deleteAtIndex.
729	★	我的日程安排表 I		Book time intervals on a calendar; return false if overlap exists.
731	★	我的日程安排表 II		Book events; return number of existing bookings overlapping the new booking.
732	★	My Calendar III		Book events; return the maximum k-booking (k events overlap at same time).
745		#745	O(L) / O(n)	Prefix and suffix search --- find word with given prefix and suffix.
933	★	Number of Recent Calls		Find the number of requests made in the last 3000 milliseconds.
1570	★	Dot Product of Two Sparse Vectors		Compute the dot product of two sparse vectors efficiently.

Math (57 problems, 57 templated)

#	★	Name	Complexity	Description
9	★	Palindrome Number	O(log n) / O(1)	Determine whether an integer reads the same forward and backward. Negative numbers are never palindromes.
29	★	Divide Two Integers		Divide two integers without using multiplication, division, or mod. Clamp to 32-bit signed range.
50	★	Pow(x, n)	O(log n) / O(1)	Implement pow(x, n) --- raise x to the power n (n can be negative).
59	★	Spiral Matrix II		Given `n`, generate an `n x n` matrix filled with the numbers `1` to `n*n` in spiral order.
66	★	Plus One		Increment a large integer represented as an array of digits.
118	★	Pascal's Triangle		Generate Pascal's triangle up to numRows rows.
119	★	Pascal's Triangle II		Return the kth row of Pascal's triangle using O(k) space.
149	★	Max Points on a Line		Find the maximum number of points that lie on the same straight line.
172	★	Factorial Trailing Zeroes		Count trailing zeroes in n! (count factors of 5).
202	★	Happy Number		Determine if a number eventually reaches 1 by repeatedly replacing it with the sum of squares of its digits.
204	★	Count Primes	O(n log log n) / O(n)	Count prime numbers less than a non-negative integer n.
223	★	Rectangle Area		Given the coordinates of two axis-aligned rectangles, return the total area they cover (counting the overlap once).
233	★	Number of Digit One		Count total occurrences of digit 1 in all integers from 1 to n.
243	★	Shortest Word Distance		Given an array of words and two distinct words, return the shortest distance (in indices) between any occurrence of the two words.
258	★	Add Digits		Repeatedly sum digits until a single digit remains (digital root).
263	★	Ugly Number		Check if a number is an ugly number (prime factors only 2, 3, 5).
292	★	Nim Game		In Nim game, you and opponent alternate removing 1--3 stones; player who takes last stone wins. Can you win?
319	★	Bulb Switcher		n bulbs toggled by n rounds (round i toggles every ith bulb). Count bulbs on after all rounds.
326	★	Power of Three		Check if a number is a power of three.
357	★	Count Numbers with Unique Digits		Given `n`, count all numbers `x` with `0 <= x < 10^n` that have no repeated digits.
365	★	Water and Jug Problem		Given two jugs of capacities `x` and `y` and a target `target`, determine if you can measure exactly `target` liters using fill/empty/pour operations.
367	★	Valid Perfect Square		Given a positive integer `num`, return whether it is a perfect square, without using any built-in sqrt function.
372	★	Super Pow	O(log n) / O(1)	Compute `a^b mod 1337` where `b` is given as an array of digits (so `b` may be astronomically large).
397	★	Integer Replacement		Find the minimum steps to reduce a number to 1 (even: n/2, odd: n+1 or n-1).
398	★	Random Pick Index		Pick a random index from those matching a target value (unknown total size).
400	★	Nth Digit		Given `n`, return the `n`-th digit in the infinite sequence of concatenated positive integers `123456789101112...`.
412	★	Fizz Buzz		Print numbers 1--n, replacing multiples of 3 with 'Fizz', 5 with 'Buzz', both with 'FizzBuzz'.

#	★	Name	Complexity	Description
441	★	Arranging Coins		You have `n` coins to form a staircase where the `k` -th row has exactly `k` coins; return the number of complete rows.
453	★	Minimum Moves to Equal Array Elements		In one move you increment `n−1` elements of the array by 1; return the minimum moves to make all elements equal.
458	★	Poor Pigs		With `buckets` buckets one of which is poisoned, `minutesToDie` for poison to act, and `minutesToTest` total time, return the minimum pigs needed to find the poisoned bucket.
470	★	Implement Rand10() Using Rand7()		Implement rand10() using only rand7().
492	★	Construct the Rectangle		Given a target `area` , return the dimensions `[L, W]` with `L >= W` , `L * W == area` , and `L − W` minimized.
495	★	Teemo Attacking		Given sorted attack `timeSeries` and a poison `duration` , return the total time the target is poisoned (overlaps counted once).
528	★	Random Pick with Weight		Randomly pick an index proportional to its weight.
593	★	Valid Square		Given four points, return whether they form a valid square (positive area).
598	★	Range Addition II		Count cells in the intersection of all rectangles defined by range addition operations.
633	★	Sum of Square Numbers		Given a non-negative integer `c` , decide whether there exist non-negative integers `a, b` with `a*a + b*b == c` .
656	★	Coin Path		Given `coins` (cost per index, `−1` blocked) and a max jump `maxJump` , find the lexicographically smallest cheapest path of indices from index 0 to the last.
781	★	Rabbits in Forest		Each surveyed rabbit reports how many other rabbits share its color; given the `answers` , return the minimum number of rabbits in the forest.
789	★	Escape The Ghosts		Starting at the origin with a `target` , and given `ghosts` positions, you all move simultaneously in Manhattan steps; return whether you can reach the target before any ghost catches you.
792	★	Number of Matching Subsequences		Given a string `s` and an array of `words` , return how many words are subsequences of `s` .
829	★	Consecutive Numbers Sum		Find the number of ways to write n as a sum of consecutive positive integers.
836	★	Rectangle Overlap		Check if two axis-aligned rectangles overlap.
858	★	Mirror Reflection		Given clock hands at h:m, find the minimum angle between them.
869	★	Reordered Power of 2		Check if any permutation of n's digits is a power of 2.
939	★	Minimum Area Rectangle		Find the minimum area rectangle from a set of points in an axis-aligned grid.
963	★	Minimum Area Rectangle II		Find the minimum area rectangle aligned to any angle from a set of points.
1041	★	Robot Bounded In Circle		Determine if a robot following instructions stays within a bounded circle.
1160	★	Find Words That Can Be Formed by Characters		Given `words` and a string `chars` , return the total length of words that can be formed using letters of `chars` (each letter used at most as often as it appears).
1201	★	Ugly Number III	O(log(min)) / O(1)	Find the nth positive integer that is divisible by at least one of `a` , `b` , or `c` .
1304	★	Find N Unique Integers Sum up to Zero		Find n unique integers that sum to zero.
1344	★	Angle Between Hands of a Clock		Calculate the angle between clock hands at a given time.
1583	★	Count Unhappy Friends		Count the number of unhappy friends after pairing.
1588	★	Sum of All Odd Length Subarrays		Compute the sum of all odd-length subarray sums.

#	★	Name	Complexity	Description
1646	★	Get Maximum in Generated Array		Generate an array from a given sequence and return the max element.
1823	★	Find the Winner of the Circular Game		`n` friends in a circle eliminate every `k` -th friend repeatedly; return the 1-indexed winner (Josephus problem).
1969	★	Minimum Non-Zero Product of the Array Elements		Find the minimum non-zero product of array elements after swaps.

Concurrency (no dedicated template) (2 problems, 0 templated)

#	★	Name	Complexity	Description
1114		Print in Order		
1115		Print FooBar Alternately		

Total: 729 problems indexed across 14 categories.